Requirements Verification

Evidence for each of the 11 project requirements, with status and screenshots.

Blake Otto · Catalyst Switch Lab — Small Office Network Build
github.com/blakeotto57/Catalyst-Switch-Lab-Small-Office-Network-Build

Revised October 10, 2026. The switch login name is blacked out in screenshots.

Requirement verification

Evidence for each requirement, taken from my documented project brief. Each status uses one of four labels:

- **Met**: the acceptance criteria are met and the evidence is below.
- **Met with limitation**: met wherever the hardware allows; the gap is a hardware limitation (see limitations.md).
- **Partially met**: part of the criteria isn't met yet.
- **Evidence pending**: the verification output isn't in this document yet.

#	Requirement	Status
1	Clean baseline	Evidence pending — `show version` and post-reset output not captured yet
2	Secure management	Met with limitation — SW3 has no SSH, so remote access to it is disabled (console only)
3	Segmentation	Partially met — SW3 Gi0/1–2 are shut down but still in VLAN 1
4	Redundant core uplinks	Met with limitation — SW3 (3548 XL) has no LACP, so its uplink is a static channel; its trunks also carry VLAN 1 and 1002–1005 and have no `switchport nonegotiate`
5	Access cross-link	Met
6	Spanning tree	Met with limitation — SW3 (3548 XL) runs 802.1D only, and its server port has PortFast but no BPDU Guard
7	Centralized routing	Met with limitation — reachability is shown from the VLAN 10 laptop on SW2 to SW3's management IP, not to the server
8	DHCP	Met with limitation — leases shown on SW2 only; SW3 has no VLAN 10 access port
9	Logging and backups	Met
10	Internet access	Partially met — lab hosts can't reach the internet (no NAT on R1, no return route on the ISP gateway)
11	Access-layer security (optional)	Partially met — met on SW2; SW3 can't run DHCP snooping and has no user ports

Requirement 1: Clean baseline

Accepted when: All three switches are inventoried (model, software, capabilities) and reset to a blank configuration

Status: Evidence pending — `show version` and post-reset output not captured yet

> Not captured yet: `show version` for each switch (model, IOS version, capabilities) and the blank configuration after the reset.

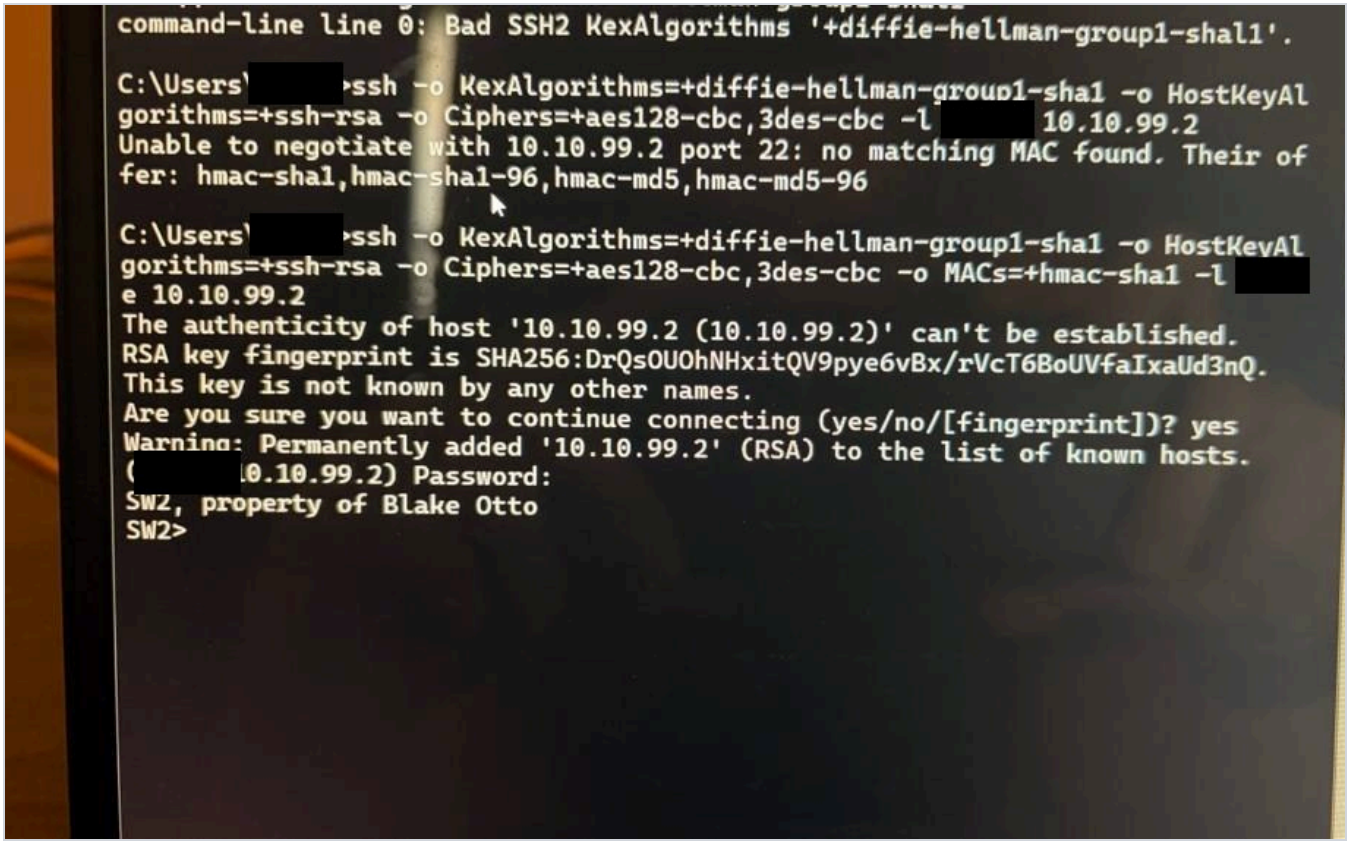
Requirement 2: Secure management

Accepted when: Every switch has a hostname, encrypted privileged password, local user login and banner; remote access is SSH version 2 only and Telnet is refused

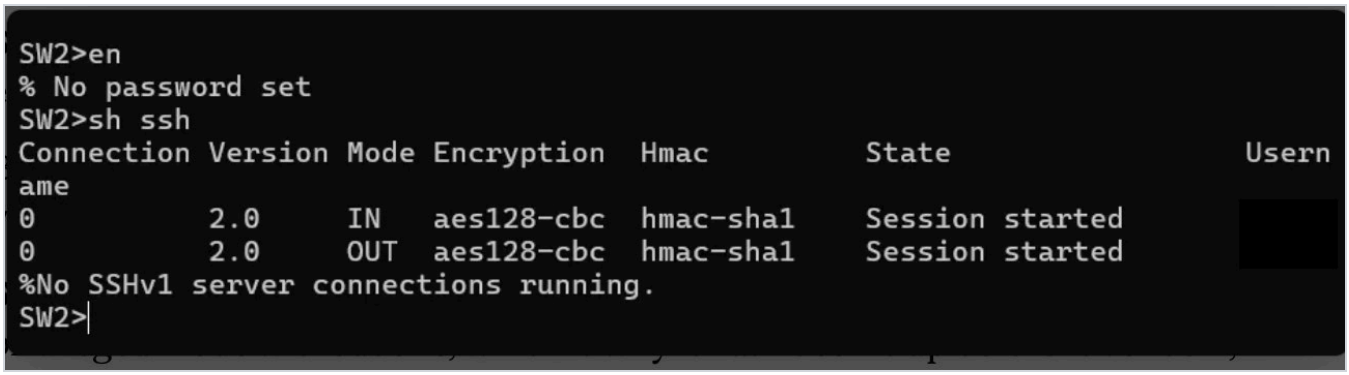
Status: Met with limitation — SW3 has no SSH, so remote access to it is disabled (console only)

Evidence and notes

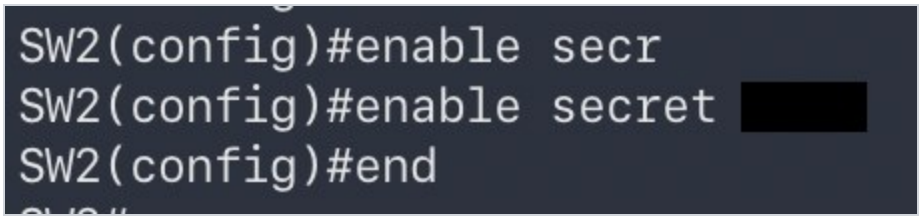
SSH to SW2 works once the laptop allows the switch's older key-exchange, host-key, cipher and MAC algorithms:



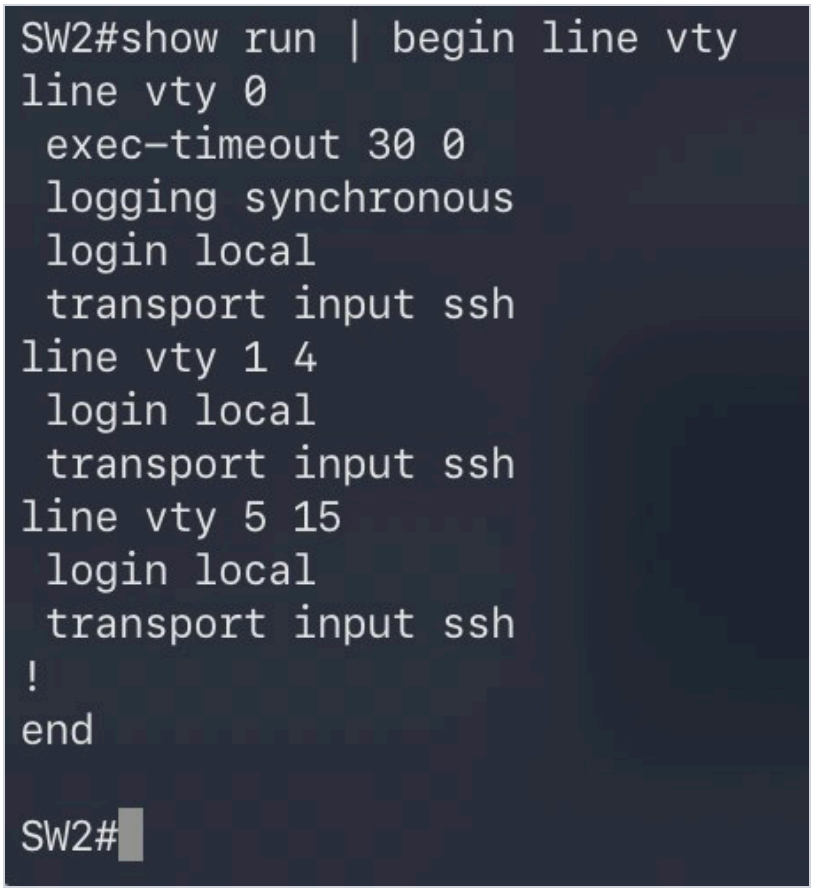
The session runs SSH version 2. This capture is from before I set an enable secret, which is why `enable` returns "% No password set":



I then set the enable secret on SW2 (value hidden):



Telnet still worked at first, because I had configured `line vty 0 4` instead of `line vty 0 15`. The corrected vty lines on SW2 and R1:



```
R1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#line vty 0 15
R1(config-line)#trans
R1(config-line)#transport in
R1(config-line)#transport input ssh
R1(config-line)#login local
R1(config-line)#end
R1#s ru
*Mar  1 14:35:48.779: %SYS-5-CONFIG_I: Configured from console by [redacted] on console
R1#sh run | begin line vty
line vty 0
 exec-timeout 30 0
 logging synchronous
 login local
 transport input ssh
line vty 1 4
 login local
 transport input ssh
line vty 5 15
 login local
 transport input ssh
!
end

R1#
```

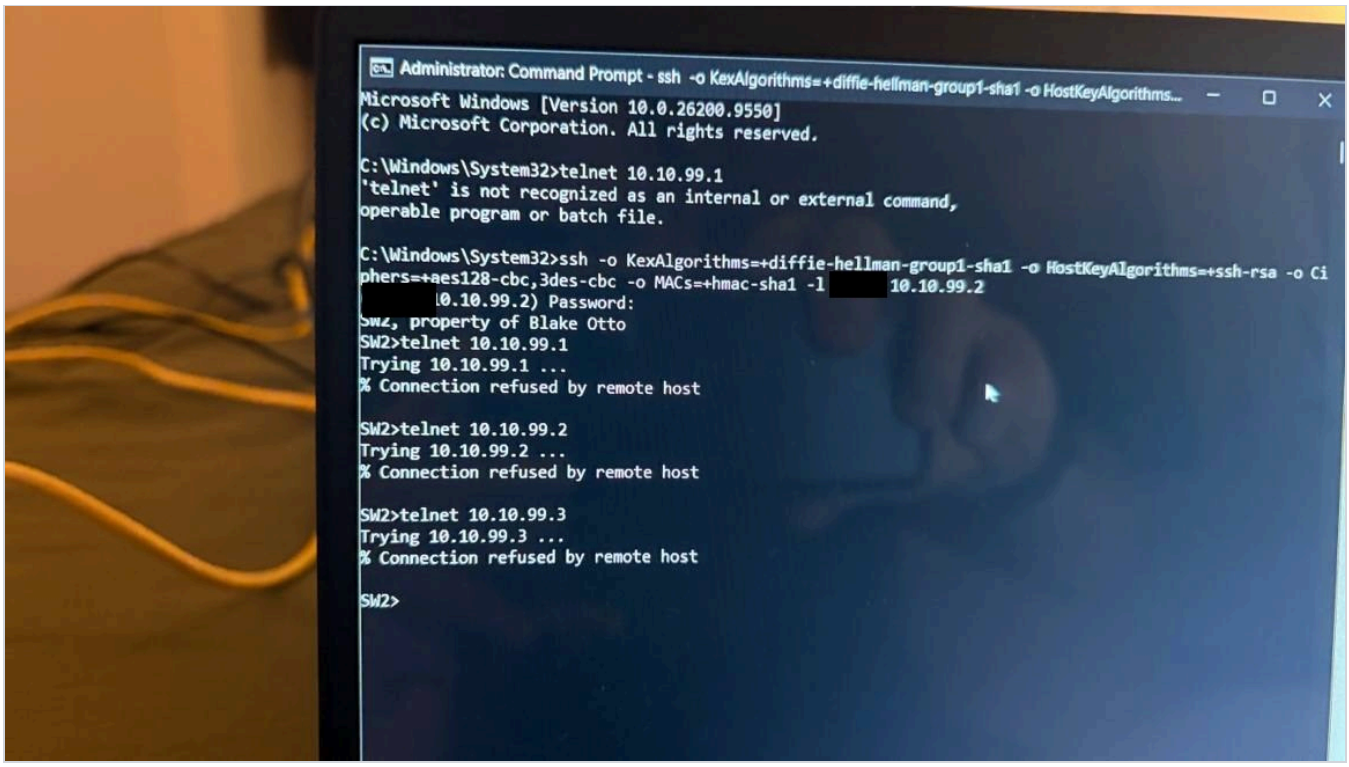
SW3's IOS has no SSH (`transport input` offers only telnet, all and none), so I disabled remote access to it entirely and manage it from the console:

```
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#line vty 0 15
SW3(config-line)#transport input ssh
                                  ^
% Invalid input detected at '^' marker.

SW3(config-line)#transport input ?
  all     All protocols
  none    No protocols
  telnet  TCP/IP Telnet protocol

SW3(config-line)#transport input none
SW3(config-line)#login local
SW3(config-line)#end
SW3#c
```

Telnet is refused by all three switches:



Every switch has an enable secret, a local user and a banner. The sanitized configs show the lines with the values removed (`enable secret <removed>`, `username <removed> …`, `banner motd`). R1's SSH login isn't in a screenshot; its config has `ip ssh version 2` and `transport input ssh` on all vty lines.

Requirement 3: Segmentation

Accepted when: VLANs 10, 20 and 99 exist on all three switches; every unused port is in VLAN 999 and shut down

Status: Partially met — SW3 Gi0/1–2 are shut down but still in VLAN 1

Evidence and notes

> These captures were taken early in the build, before the trunks and host ports were configured (for example, SW2 Gi0/3 is still in VLAN 1 and SW3 Fa0/3–4 are still in VLAN 999). They show that VLANs 10, 20, 99 and 999 exist on all three switches. They also show SW3's Gi0/1–2 in VLAN 1, which is still true in the final config. `show vlan` doesn't show whether ports are shut down; the sanitized configs do.

```
SW2#show vlan

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gi0/1, Gi0/2, Gi0/3, Gi0/4
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Gi0/5, Gi0/6, Gi0/7, Gi0/8
                                                Gi0/9, Gi0/10, Gi0/11, Gi0/12
                                                Gi0/13, Gi0/14, Gi0/15, Gi0/16
                                                Gi0/17, Gi0/18, Gi0/19, Gi0/20
                                                Gi0/21, Gi0/22, Gi0/23, Gi0/24
                                                Gi0/25, Gi0/26, Gi0/27, Gi0/28
                                                Gi0/29, Gi0/30, Gi0/31, Gi0/32
                                                Gi0/33, Gi0/34, Gi0/35, Gi0/36
                                                Gi0/37, Gi0/38, Gi0/39, Gi0/40
                                                Gi0/41, Gi0/42, Gi0/43, Gi0/44
                                                Gi0/45, Gi0/46, Gi0/47, Gi0/48
                                                Gi0/49, Gi0/50, Gi0/51, Gi0/52
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
1005 trnet-default                    act/unsup
 --More--
```

```
SW3#show vlan
VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Gi0/1, Gi0/2
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Fa0/3, Fa0/4, Fa0/5, Fa0/6,
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10,
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14,
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18,
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22,
                                                Fa0/23, Fa0/24, Fa0/25, Fa0/26,
                                                Fa0/27, Fa0/28, Fa0/29, Fa0/30,
                                                Fa0/31, Fa0/32, Fa0/33, Fa0/34,
                                                Fa0/35, Fa0/36, Fa0/37, Fa0/38,
                                                Fa0/39, Fa0/40, Fa0/41, Fa0/42,
                                                Fa0/43, Fa0/44, Fa0/45, Fa0/46,
                                                Fa0/47, Fa0/48
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
R1(config)#do sh vlan br

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Gi0/1, Gi0/2, Gi0/3
                                                Gi0/4
10   USERS                            active
20   SERVERS                          active
99   MGMT                             active
999  UNUSED                           active    Fa0/2, Fa0/3, Fa0/4, Fa0/5
                                                Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Fa0/25
                                                Fa0/26, Fa0/27, Fa0/28, Fa0/29
                                                Fa0/30, Fa0/31, Fa0/32, Fa0/33
                                                Fa0/34, Fa0/35, Fa0/36, Fa0/37
                                                Fa0/38, Fa0/39, Fa0/40, Fa0/41
                                                Fa0/42, Fa0/43, Fa0/44, Fa0/45
                                                Fa0/46, Fa0/47, Fa0/48
1002 fddi-default                     act/unsup
1003 token-ring-default               act/unsup
1004 fddinet-default                  act/unsup
 --More--
```

Requirement 4: Redundant core uplinks

Accepted when: Each access switch connects to the core through its own two-link LACP EtherChannel; each trunk carries only lab VLANs, uses VLAN 900 as native and does not negotiate trunking

Status: Met with limitation — SW3 (3548 XL) has no LACP, so its uplink is a static channel; its trunks also carry VLAN 1 and 1002–1005 and have no `switchport nonegotiate`

Evidence and notes

EtherChannel summaries from failure test 1. SW2's Po1 runs LACP with both links bundled. On R1, Po1 (to SW2) runs LACP and Po2 (to SW3) is a static channel, shown with protocol "-":

```
SW2#sh ether sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 1
Number of aggregators:           1

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-----------------------------------
1      Po1(SU)         LACP      Gi0/1(P)    Gi0/2(P)

SW2#
```

```
R1#sh eth sum
Flags:  D - down        P - bundled in port-channel
        I - stand-alone s - suspended
        H - Hot-standby (LACP only)
        R - Layer3      S - Layer2
        U - in use      f - failed to allocate aggregator

        M - not in use, minimum links not met
        u - unsuitable for bundling
        w - waiting to be aggregated
        d - default port


Number of channel-groups in use: 2
Number of aggregators:           2

Group  Port-channel  Protocol    Ports
------+-------------+-----------+-------------------------------
1      Po1(SU)         LACP      Fa0/1(P)    Fa0/2(P)
2      Po2(SU)          -        Fa0/3(P)    Fa0/4(P)

R1#
```

Trunk settings from the sanitized configs:

```
R1: Port-channel1, Port-channel2 and their member ports (Fa0/1–4)
SW2: Port-channel1, Gi0/1–2 and the cross-link Gi0/4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 900
 switchport trunk allowed vlan 10,20,99,900
 switchport mode trunk
 switchport nonegotiate

SW3: Fa0/1–2 (port group 2) and the cross-link Fa0/4
 switchport trunk encapsulation dot1q
 switchport trunk native vlan 900
 switchport trunk allowed vlan 1,10,20,99,900,1002-1005
 switchport mode trunk
```

SW3 (3548 XL) can't remove VLAN 1 and 1002–1005 from a trunk, and its trunk config has no `switchport nonegotiate`. R1 and SW2 don't allow those VLANs, so they never cross the links. A `show interfaces trunk` capture isn't included yet.

Requirement 5: Access cross-link

Accepted when: A single trunk between SW2 and SW3 provides a backup path to the core

Status: Met

Evidence and notes

SW2's end of the cross-link (Gi0/4) is designated forwarding, and its root port is Po1 toward R1:

```
SW2#show spanning-tree vlan 10

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p


SW2#
```

SW3's end of the cross-link (Fa0/4) is blocking, and its root port is the uplink toward R1. The link is in place as a backup path; failure test 2 shows it taking over.

```
SW3#show spanning-tree vlan 10

Spanning tree 10 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6201
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24586, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 3
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 10 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 24586, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 2, forward delay 0, hold 0
   BPDU: sent 1, received 4004

Interface Fa0/4 (port 16) in Spanning tree 10 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24586, address 0016.9d94.0980
   Designated bridge has priority 28682, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 11, forward delay 0, hold 0
   BPDU: sent 1, received 875
SW3#
```

Requirement 6: Spanning tree

Accepted when: Rapid PVST+ runs everywhere; R1 is root for every lab VLAN and SW2 is secondary root; the cross-link is blocked on SW3's side; edge ports use PortFast and BPDU Guard

Status: Met with limitation — SW3 (3548 XL) runs 802.1D only, and its server port has PortFast but no BPDU Guard

Evidence and notes

SW2 runs Rapid PVST+, but this first check showed SW2 as root for VLAN 99. At that point R1's priority covered only VLANs 10 and 20:

```
SW2(config)#do show spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0099
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is enabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                    0         0        0          3          3
VLAN0020                    0         0        0          2          2
VLAN0099                    0         0        0          2          2
VLAN0900                    0         0        0          2          2
---------------------- -------- --------- -------- ---------- ----------
4 vlans                     0         0        0          9          9
```

After I set priorities for every lab VLAN, R1 runs Rapid PVST+ and is root for VLANs 10, 20, 99 and 900:

```
R1(config)#spanning-tree mode rapid-pvst
R1(config)#end
R1#sh
*Mar  1 11:47:13.921: %SYS-5-CONFIG_I: Configured from console by [redacted] on console
R1#sh spann
R1#sh spanning-tree sum
R1#sh spanning-tree summary | i mode
Switch is in rapid-pvst mode
R1#sh spann
R1#sh spanning-tree sum
R1#sh spanning-tree summary
Switch is in rapid-pvst mode
Root bridge for: VLAN0010, VLAN0020, VLAN0099, VLAN0900
Extended system ID           is enabled
Portfast Default             is disabled
PortFast BPDU Guard Default  is disabled
Portfast BPDU Filter Default is disabled
Loopguard Default            is disabled
EtherChannel misconfig guard is enabled
UplinkFast                   is disabled
BackboneFast                 is disabled
Configured Pathcost method used is short

Name                   Blocking Listening Learning Forwarding STP Active
---------------------- -------- --------- -------- ---------- ----------
VLAN0010                    0         0        0          2          2
VLAN0020                    0         0        0          2          2
VLAN0099                    0         0        0          2          2
VLAN0900                    0         0        0          2          2
---------------------- -------- --------- -------- ---------- ----------
4 vlans                     0         0        0          8          8
R1#
```

SW2 is the secondary root (priority 28672) with Po1 as its root port, and its user port Gi0/3 is an edge port:

```
SW2#sh span

VLAN0010
  Spanning tree enabled protocol rstp
  Root ID    Priority    24586
             Address     0016.9d94.0980
             Cost        12
             Port        56 (Port-channel1)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28682  (priority 28672 sys-id-ext 10)
             Address     0019.e885.9000
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- --------------------------------
Gi0/3               Desg FWD 4         128.3    P2p Edge
Gi0/4               Desg FWD 19        128.4    P2p Peer(STP)
Po1                 Root FWD 12        128.56   P2p
```

SW3 can't run Rapid PVST+. It still sees R1 as root for VLAN 99 and blocks the cross-link on Fa0/4:

```
SW3#show spanning-tree vlan 99

Spanning tree 99 is executing the IEEE compatible Spanning Tree protocol
  Bridge Identifier has priority 32768, address 0007.0ecc.6203
  Configured hello time 2, max age 20, forward delay 15
  Current root has priority 24675, address 0016.9d94.0980
  Root port is 2, cost of root path is 12
  Topology change flag not set, detected flag not set, changes 6
  Times:  hold 1, topology change 35, notification 2
          hello 2, max age 20, forward delay 15
  Timers: hello 0, topology change 0, notification 0

Interface Fa0/2 (port 2) in Spanning tree 99 is FORWARDING
   Port path cost 12, Port priority 128
   Designated root has priority 24675, address 0016.9d94.0980
   Designated bridge has priority 24675, address 0016.9d94.0980
   Designated port is 64, path cost 0
   Timers: message age 1, forward delay 0, hold 0
   BPDU: sent 76, received 5916

Interface Fa0/4 (port 16) in Spanning tree 99 is BLOCKING
   Port path cost 19, Port priority 128
   Designated root has priority 24675, address 0016.9d94.0980
   Designated bridge has priority 28771, address 0019.e885.9000
   Designated port is 4, path cost 12
   Timers: message age 4, forward delay 0, hold 0
   BPDU: sent 4, received 2853
SW3#
```

Edge ports: SW2's user port Gi0/3 has PortFast and BPDU Guard (failure test 5 shows BPDU Guard working). SW3's server port Fa0/3 has PortFast only; see SW3-sanitized.txt.

```
SW2#
SW2#sh run int g0/3 | i portfast|bpdu
 spanning-tree portfast
 spanning-tree bpduguard enable
SW2#sh spann
SW2#sh spanning-tree int g0/3 port
VLAN0010             enabled
SW2#
```

Requirement 7: Centralized routing

Accepted when: R1 is the only routing device and the gateway for every VLAN; SW2 and SW3 have only a management address and a default gateway; a VLAN 10 device on SW2 reaches the server on SW3

Status: Met with limitation — reachability is shown from the VLAN 10 laptop on SW2 to SW3's management IP, not to the server

> The test laptop and the server VM share one USB network adapter, so the laptop can't be on SW2 while the server is on SW3. SW3's MAC table shows both MACs on Fa0/3 (requirement 10).

Evidence and notes

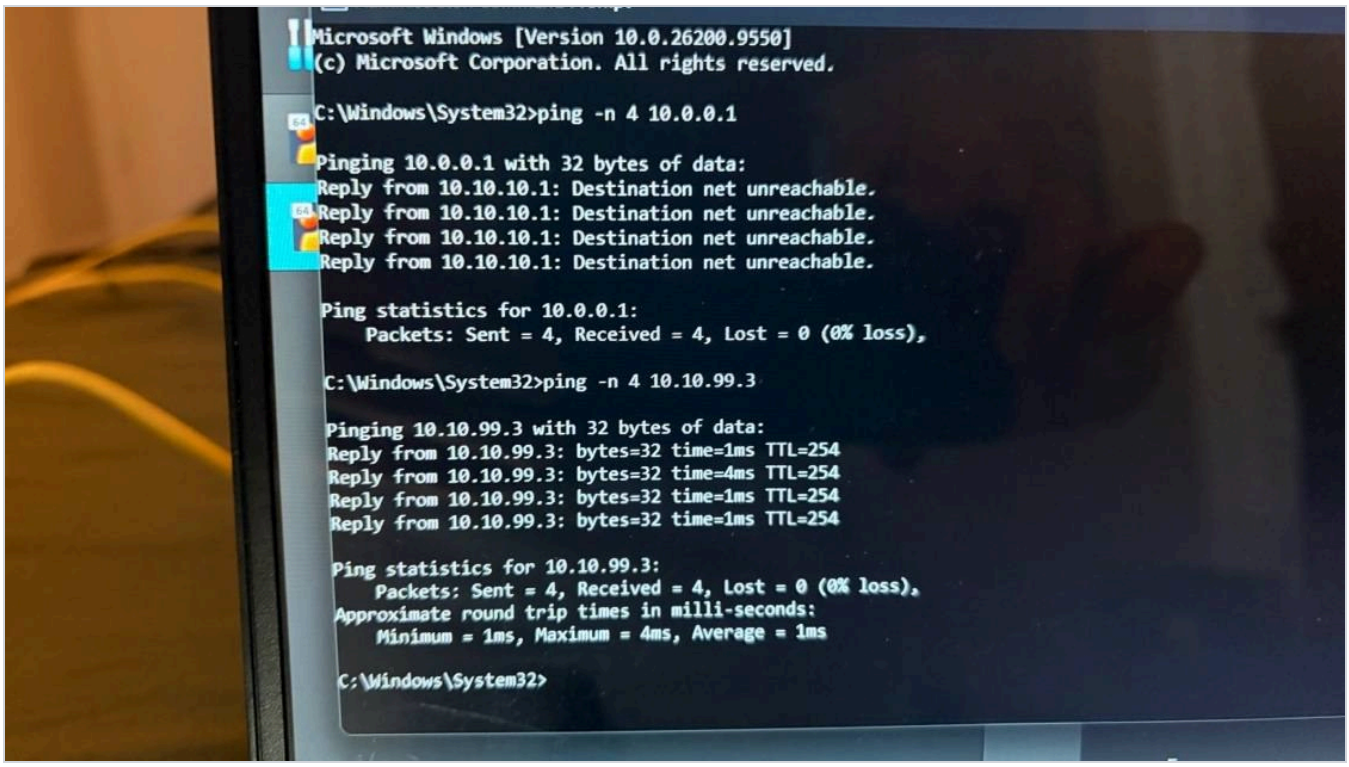
Routing happens only on R1. From the sanitized configs:

```
R1-sanitized.txt
 ip routing
 interface Vlan10
  ip address 10.10.10.1 255.255.255.0
 interface Vlan20
  ip address 10.10.20.1 255.255.255.0
 interface Vlan99
  ip address 10.10.99.1 255.255.255.0
 ip route 0.0.0.0 0.0.0.0 10.0.0.1

SW2-sanitized.txt
 interface Vlan99
  ip address 10.10.99.2 255.255.255.0
 ip default-gateway 10.10.99.1

SW3-sanitized.txt
 interface VLAN99
  ip address 10.10.99.3 255.255.255.0
 ip default-gateway 10.10.99.1
```

The laptop in VLAN 10 on SW2 reaches SW3's management address in VLAN 99 through R1 (this capture is from failure test 8, so the ping to the home router is blocked by the ACL):



A `show ip route` capture from R1 isn't included yet.

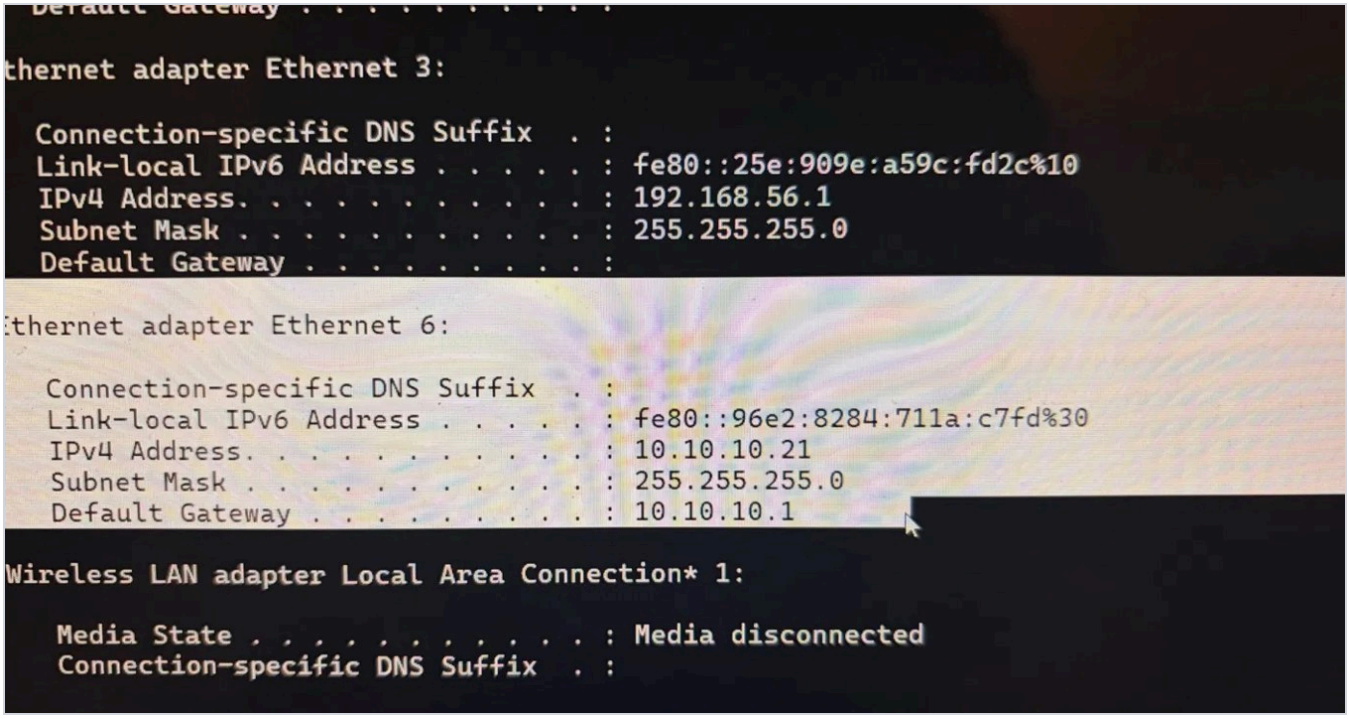
Requirement 8: DHCP

Accepted when: R1 assigns VLAN 10 addresses to devices on either access switch, reserving .1 through .20, with the correct gateway

Status: Met with limitation — leases shown on SW2 only; SW3 has no VLAN 10 access port

Evidence and notes

With the laptop on SW2 Gi0/3, it got 10.10.10.21 with gateway 10.10.10.1, the first address after the excluded range .1–.20:



SW2's DHCP snooping table records that lease on Gi0/3 in VLAN 10:

```
SW2#show port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : c8a3.62a2.1bed:10
Security Violation Count   : 0

SW2#sh ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type           VLAN  Interface
------------------  ---------------  ----------  -------------  ----  --------------------
C8:A3:62:A2:1B:ED   10.10.10.21      86291       dhcp-snooping   10    GigabitEthernet0/3
Total number of bindings: 1

SW2#show run int g0/3 | i sticky
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky c8a3.62a2.1bed
SW2#
```

R1's DHCP configuration, from R1-sanitized.txt:

```
ip dhcp excluded-address 10.10.10.1 10.10.10.20
ip dhcp pool USERS
   network 10.10.10.0 255.255.255.0
   default-router 10.10.10.1
   dns-server 8.8.8.8
```

Not shown: a lease on SW3 (it has no VLAN 10 access port) and R1's `show ip dhcp binding`.

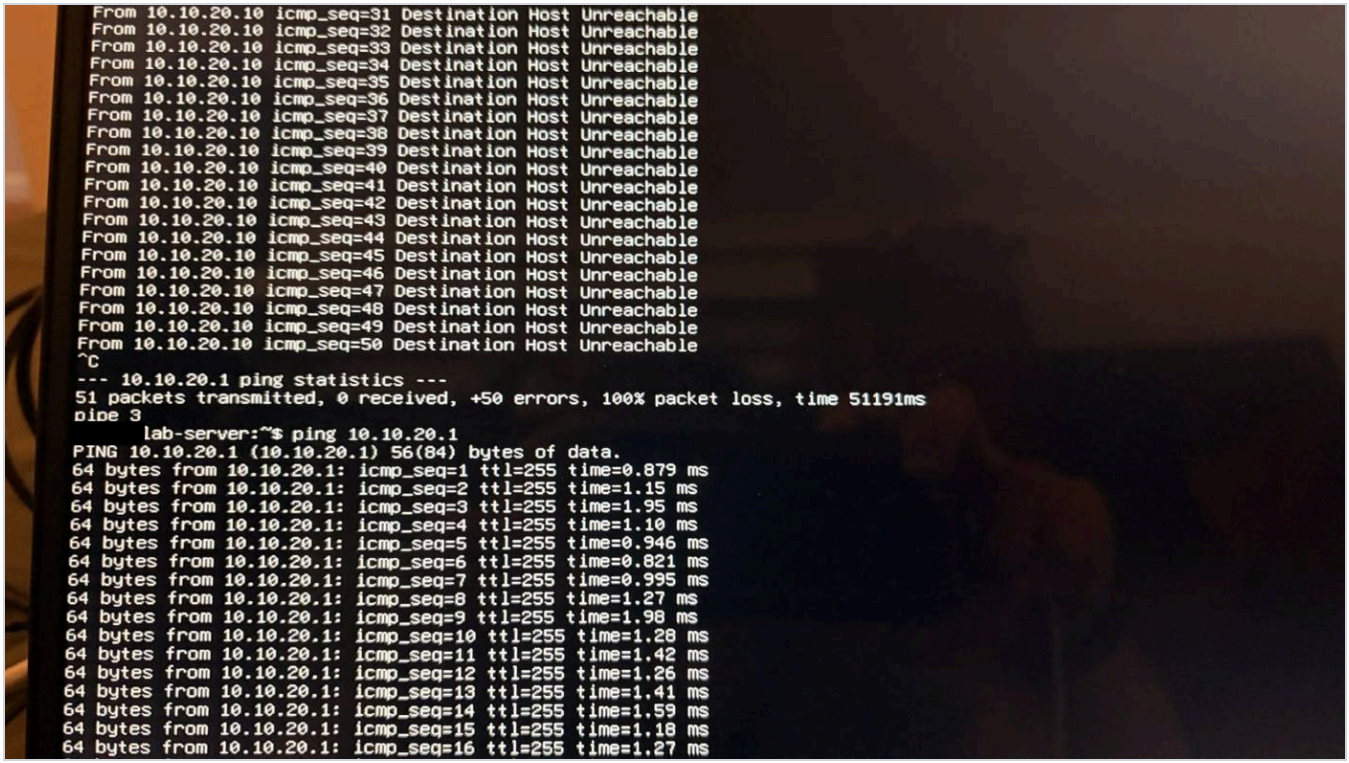
Requirement 9: Logging and backups

Accepted when: The Ubuntu server receives timestamped syslog from all three switches and stores all three running configurations

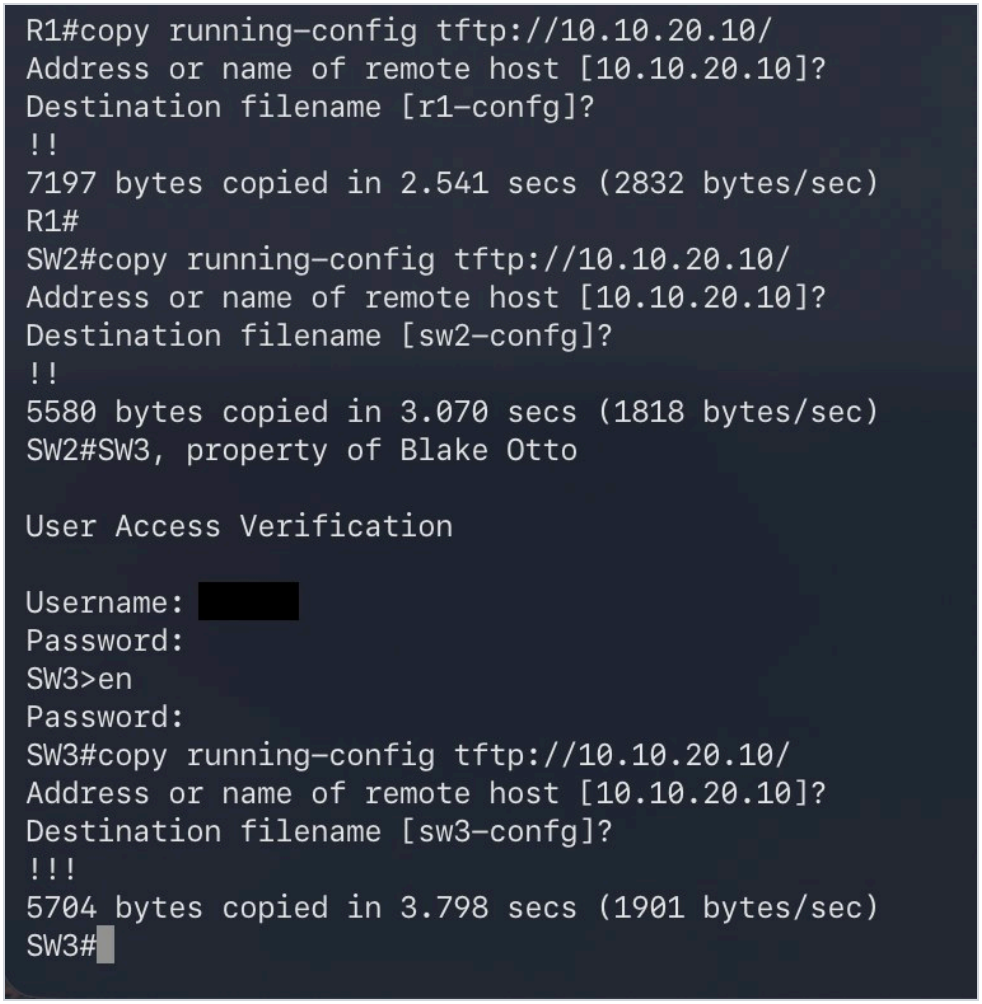
Status: Met

Evidence and notes

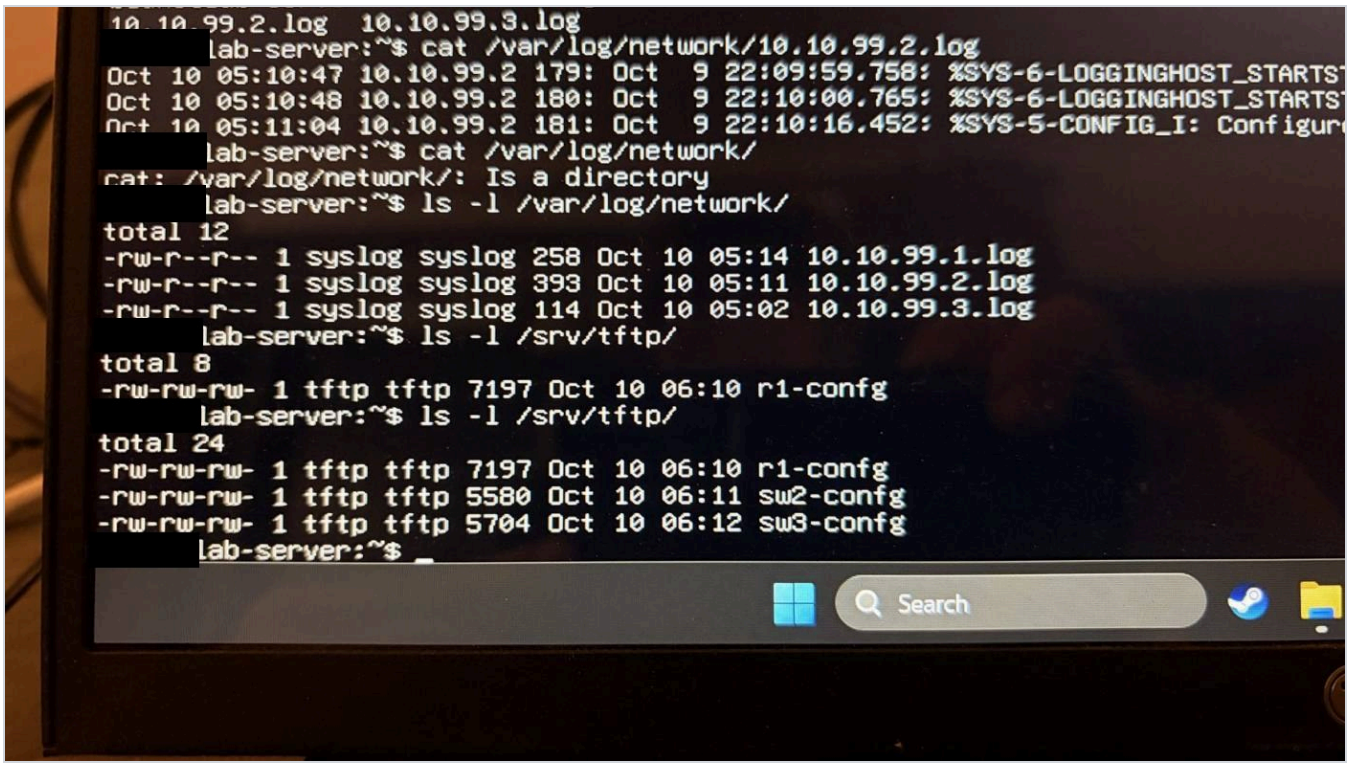
Before logging could work, SRV1 had to reach its gateway. The first attempts failed until I restarted the VM and its bridged USB adapter (troubleshooting log #1):



I saved the running configs of all three switches to SRV1 over TFTP:



SRV1 keeps a syslog file per switch (10.10.99.1, .2 and .3) and the three config backups. The SW2 log is shown; the other two appear in the listing. Device timestamps and SRV1's timestamps differ because the lab has no NTP (see limitations.md).


10.10.99.2.log 10.10.99.3.log
lab-server:~$ cat /var/log/network/10.10.99.2.log
Oct 10 05:10:47 10.10.99.2 179: Oct 9 22:09:59.758: %SYS-6-LOGGINGHOST_STARTST
Oct 10 05:10:48 10.10.99.2 180: Oct 9 22:10:00.765: %SYS-6-LOGGINGHOST_STARTST
Oct 10 05:11:04 10.10.99.2 181: Oct 9 22:10:16.452: %SYS-5-CONFIG_I: Configur
lab-server:~$ cat /var/log/network/
cat: /var/log/network/: Is a directory
lab-server:~$ ls -l /var/log/network/
total 12
-rw-r--r-- 1 syslog syslog 258 Oct 10 05:14 10.10.99.1.log
-rw-r--r-- 1 syslog syslog 393 Oct 10 05:11 10.10.99.2.log
-rw-r--r-- 1 syslog syslog 114 Oct 10 05:02 10.10.99.3.log
lab-server:~$ ls -l /srv/tftp/
total 8
-rw-rw-rw- 1 tftp tftp 7197 Oct 10 06:10 r1-confg
lab-server:~$ ls -l /srv/tftp/
total 24
-rw-rw-rw- 1 tftp tftp 7197 Oct 10 06:10 r1-confg
-rw-rw-rw- 1 tftp tftp 5580 Oct 10 06:11 sw2-confg
-rw-rw-rw- 1 tftp tftp 5704 Oct 10 06:12 sw3-confg
lab-server:~$
Search

Requirement 10: Internet access

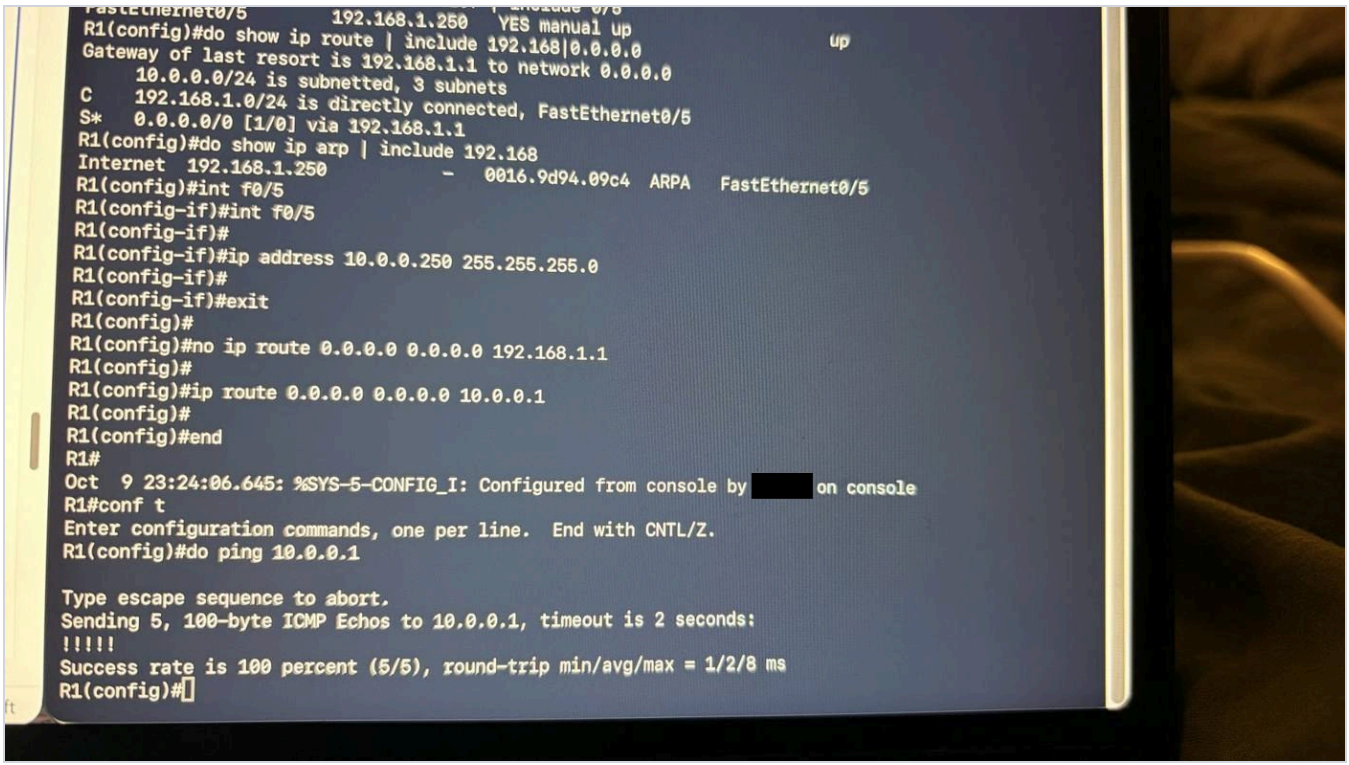
Accepted when: Lab devices reach the internet through the home router over a routed port on R1; lab VLANs and DHCP stay off the home network, and an access list blocks lab devices from home devices

Status: Partially met — lab hosts can't reach the internet (no NAT on R1, no return route on the ISP gateway)

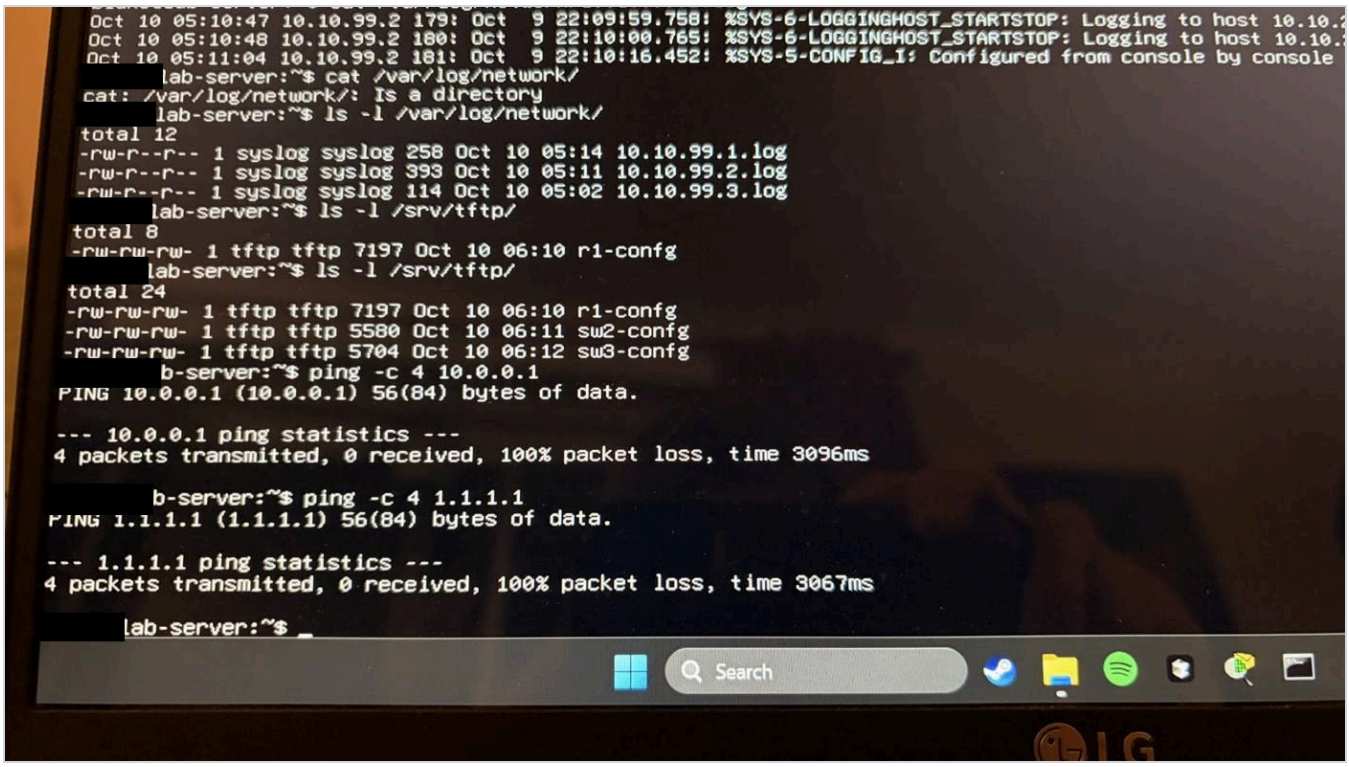
> R1 reaches the home router over its routed uplink, and the ACL that blocks lab devices from home devices is in place and tested. Lab hosts can't reach the internet: the core switch (WS-C3560-48TS) has no NAT, and the ISP gateway has no route back to the lab subnets, so return traffic has no path. In production, an edge router or firewall doing NAT would fix this.

Evidence and notes

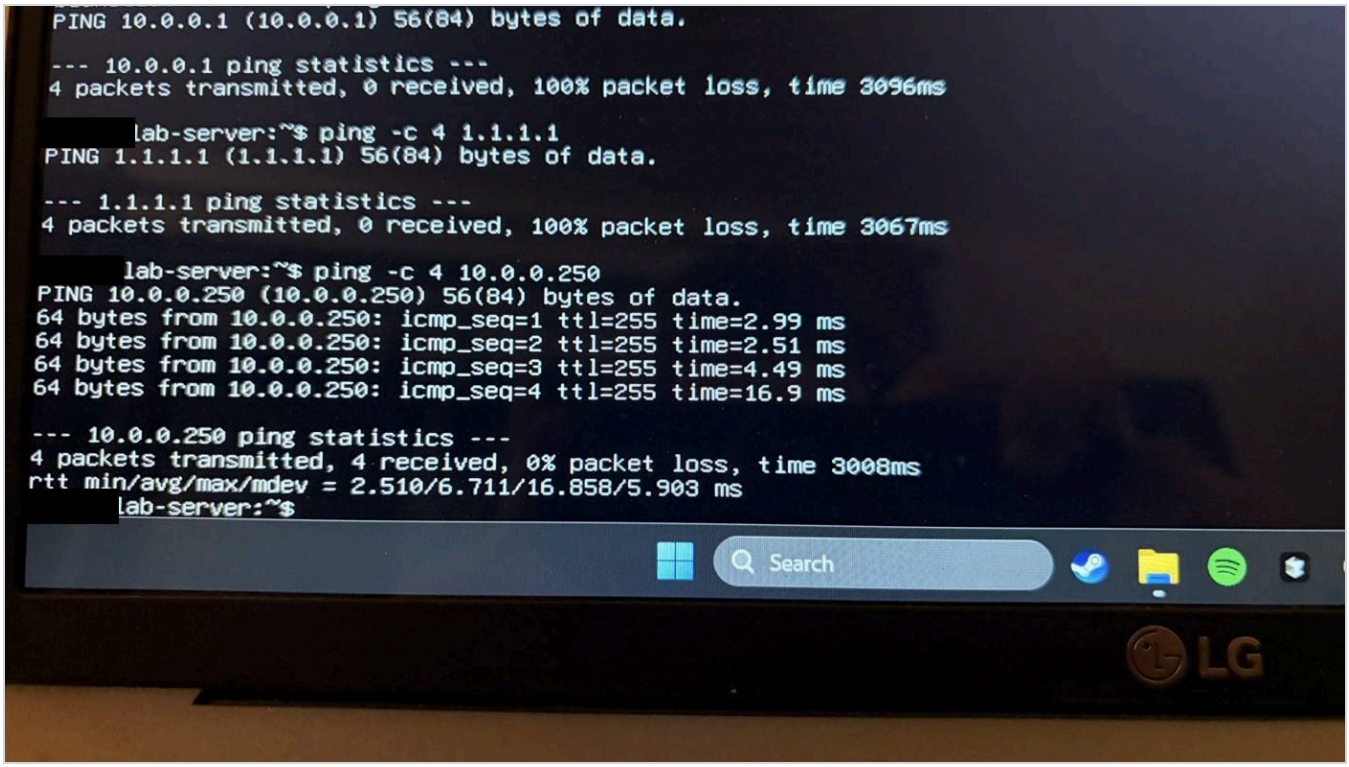
R1's uplink Fa0/5 moved to the home network (10.0.0.250/24) with a default route to the home router, and R1 can ping the router at 10.0.0.1:



SRV1 can't reach the home router (blocked by the ACL) or the internet (1.1.1.1, no return path):



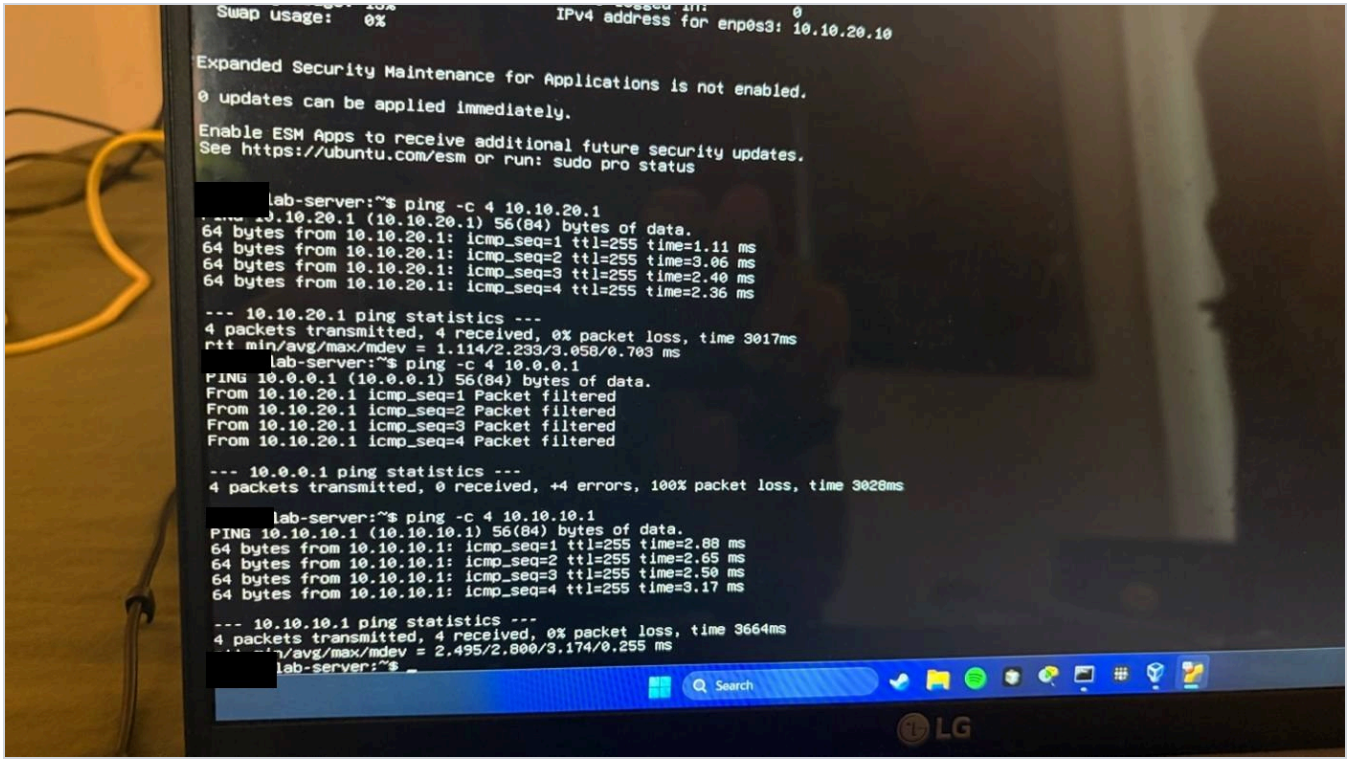
but it can reach R1's uplink address, 10.0.0.250:



SW3's MAC table on Fa0/3 shows the server VM's MAC (0800.2749.db1d) and the laptop's MAC (c8a3.62a2.1bed), because both share one USB adapter:

```
SW3#show mac-address-table int f0/3
Non-static Address Table:
Destination Address  Address Type  VLAN  Destination Port
-------------------  ------------  ----  --------------------
0800.2749.db1d       Dynamic         20  FastEthernet0/3
c8a3.62a2.1bed       Dynamic         20  FastEthernet0/3
SW3#
```

The ACL at work: SRV1's ping to 10.0.0.1 comes back "Packet filtered", and R1 counts and logs the denied packets on line 20 (VLAN 20):




```
R1#   show access-lists LAB_BLOCK_HOME
Extended IP access list LAB_BLOCK_HOME
    10 deny ip 10.10.10.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    20 deny ip 10.10.20.0 0.0.0.255 10.0.0.0 0.0.0.255 log (8 matches)
    30 deny ip 10.10.99.0 0.0.0.255 10.0.0.0 0.0.0.255 log
    40 permit ip any any
R1#   show logging | i IPACCESSLOG
*Mar  1 09:41:37.537: %SEC-6-IPACCESSLOGDP: list LAB_BLOCK_HOME denied icmp 10.10.20.10 -> 10.0.0.1
 (0/0), 1 packet
R1#
```

Requirement 11: Access-layer security (optional)

Accepted when: User ports on both access switches allow one learned device, and DHCP snooping protects VLAN 10 on both, trusting only the paths toward R1

Status: Partially met — met on SW2; SW3 can't run DHCP snooping and has no user ports

Evidence and notes

DHCP snooping on SW2 protects VLAN 10. It trusts the uplinks to R1 (Gi0/1, Gi0/2, Po1) and the cross-link Gi0/4, so DHCP still works if SW2 loses both uplinks (see design-decisions.md). Port security on Gi0/3 allows one MAC, with violation mode shutdown:

```
SW2#show ip dhcp snooping
Switch DHCP snooping is enabled
DHCP snooping is configured on following VLANs:
10
DHCP snooping is operational on following VLANs:
10
DHCP snooping is configured on the following L3 Interfaces:

Insertion of option 82 is disabled
   circuit-id format: vlan-mod-port
    remote-id format: MAC
Option 82 on untrusted port is not allowed
Verification of hwaddr field is enabled
Verification of giaddr field is enabled
DHCP snooping trust/rate is configured on the following Interfaces:

Interface                  Trusted     Rate limit (pps)
------------------------   -------     ----------------
GigabitEthernet0/1         yes         unlimited
GigabitEthernet0/2         yes         unlimited
GigabitEthernet0/4         yes         unlimited
Port-channel1              yes         unlimited
SW2#show port-security interface g0/3
Port Security              : Enabled
Port Status                : Secure-down
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 0
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 0
Last Source Address:Vlan   : 0000.0000.0000:0
Security Violation Count   : 0

SW2#
```

With the laptop plugged into Gi0/3, it got a lease, port security learned its MAC as sticky, and snooping recorded the binding:

```
SW2#show port-security int g0/3
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Shutdown
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 1
Total MAC Addresses        : 1
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 1
Last Source Address:Vlan   : c8a3.62a2.1bed:10
Security Violation Count   : 0

SW2#sh ip dhcp snooping binding
MacAddress          IpAddress        Lease(sec)  Type           VLAN  Interface
------------------  ---------------  ----------  -------------  ----  --------------------
C8:A3:62:A2:1B:ED   10.10.10.21      86291       dhcp-snooping   10    GigabitEthernet0/3
Total number of bindings: 1

SW2#show run int g0/3 | i sticky
 switchport port-security mac-address sticky
 switchport port-security mac-address sticky c8a3.62a2.1bed
SW2#
```

SW3's IOS doesn't have DHCP snooping:

